

- 6 (a) Fig. 6.1 is a list of infectious diseases. Each of the statements **A** to **D** describes a feature that applies to one or more of these diseases.

cholera
HIV/AIDS
malaria
measles
smallpox
tuberculosis (TB)

Fig. 6.1

For each of the statements **A**, **B**, **C** and **D**, name **all** the diseases in Fig. 6.1 that match the feature described.

- A** The causative organism of the disease is a virus.

.....

- B** The causative organism of the disease is a prokaryote.

.....

- C** The disease is transmitted by a faecal-oral route, for example, sewage containing the pathogen contaminates drinking water.

.....

- D** The causative organism of the disease spends part of its life cycle inside an insect, which acts as a vector of the disease.

.....

- (b) Although many infectious diseases are caused by prokaryotic organisms, there are some that are caused by eukaryotic organisms.

Complete Table 6.1 to show some differences between a prokaryotic cell and a eukaryotic cell.

Table 6.1

prokaryotic cell	eukaryotic cell
no true nucleus, genetic material not enclosed	true nucleus, genetic material enclosed by a double membrane known as a
..... DNA	linear DNA
70S ribosomes only	70S and ribosomes
no double membrane-bound organelles	double membrane-bound organelles such as
cell wall contains	where cell wall is present, generally contains mainly cellulose or chitin

[2]

[Total: 6]